1.

Using $\theta^{(t)}$, get conditional expectation of y_4 and y_5.

$$f_4 = \frac{1}{2} \quad f_5 = \frac{1}{4}\theta$$

$$\pi_4 = \frac{1}{2} + \frac{1}{4}\theta$$

Let's be p is the probability of being y_5

$$P = \frac{f_5}{f_4 + f_5} = \frac{\frac{1}{4}\theta^{(t)}}{\frac{1}{2} + \frac{1}{4}\theta^{(t)}} = \frac{\theta^{(t)}}{2 + \theta^{(t)}}$$

$$E[Y_5 \mid X, \theta^{(t)}] = np = x_4 \cdot \frac{\theta^{(t)}}{2+\theta^{(t)}}$$

Let's be q is the probability of being y_4.

$$q = \frac{f_4}{f_4 + f_5} = \frac{\frac{1}{2}}{\frac{1}{2} + \frac{1}{4}\theta^{(t)}} = \frac{2}{2+\theta^{(t)}}$$

$$E[Y_4 \mid X, \theta^{(t)}] = nq = x_4 \cdot \frac{2}{2+\theta^{(t)}}$$

$$P(Y_1 = y_1, \cdots Y_5 = y_5) = \frac{n!}{y_1!\, y_2!\, y_3!\, y_4!\, y_5!} \cdot f_1^{y_1} f_2^{y_2} f_3^{y_3} f_4^{y_4} f_5^{y_5}$$

$$L(\theta : y) = \frac{n!}{y_1!\, y_2!\, y_3!\, y_4!\, y_5!} \times \left(\frac{\theta}{4}\right)^{y_1} \left(\frac{1-\theta}{4}\right)^{y_2} \left(\frac{1-\theta}{4}\right)^{y_3} \left(\frac{1}{2}\right)^{y_4} \left(\frac{\theta}{4}\right)^{y_5}$$

$$\log L(\theta : y) = \log \frac{n!}{y_1!\, y_2!\, y_3!\, y_4!\, y_5!} + y_1(\log\theta) + y_5(\log\theta) + (y_2 + y_3)\log(1-\theta)$$

$$+ y_4 \log\left(\frac{1}{2}\right)$$

$$\ell(\theta) = (y_1 + y_5)\log\theta + (y_2 + y_3)\log(1-\theta) + C''$$

M-step.

$$y_1 = x_1 \qquad y_2 = x_2 \qquad y_3 = x_3 \qquad y_5 = x_4 \frac{\theta^{(t)}}{2+\theta^{(t)}}$$

$$Q(\theta \mid \theta^{(t)}) = \left(x_1 + x_4 \cdot \frac{\theta^{(t)}}{2+\theta^{(t)}}\right)\log\theta + (x_2 + x_3)\log(1-\theta) + C''$$

$$\max Q(\theta \mid \theta^{(t)})$$

$$\frac{\partial Q}{\partial \theta} = 0$$

$$\left(x_1 + x_4 \frac{\theta^{(t)}}{2+\theta^{(t)}}\right)\frac{1}{\theta} - \frac{(x_2 + x_3)}{1-\theta} = 0$$

$$(1-\theta)\underbrace{\left(x_1 + x_4 \cdot \frac{\theta^{(t)}}{2+\theta^{(t)}}\right)}_{A} - \theta(x_2 + x_3) = 0$$

$$(1-\theta)(x_1 + A) = \theta(x_2 + x_3)$$

$$x_1 + A - \theta \cdot x_1 - \theta A = \theta \cdot x_2 + \theta x_3$$

$$x_1 + A = \theta(x_2 + x_3 + x_1 + A)$$

$$\theta^{(t+1)} = \frac{x_1 + x_4 \cdot \frac{\theta^{(t)}}{2+\theta^{(t)}}}{x_1 + x_2 + x_3 + x_4 \cdot \frac{\theta^{(t)}}{2+\theta^{(t)}}}$$

Repeat until the value is converged.

2

$M_{ij} \sim$ Bernoulli (p)

$P(M_{ij}=1) = p$

$P(M_{ij}=0) = 1-p$

$\delta_k(i,j) \mid M_{ij}=1 \sim$ Bernoulli (a_k)

$a_k = P(\delta_k(i,j)=1 \mid M_{ij}=1)$

$b_k = P(\delta_k(i,j)=1 \mid M_{ij}=0)$

$P(\delta_k = d \mid M_{ij}=1) = a_k^d (1-a_k)^{1-d}$

$P(\delta_k = d \mid M_{ij}=0) = b_k^d (1-b_k)^{1-d}$

$$P(\delta(i,j) \mid M_{ij}=1) = \prod_{k=1}^{K} a_k^{\delta_k(i,j)} (1-a_k)^{1-\delta_k(i,j)}$$

$P(M_{ij}=1, \delta(i,j)) = P(M_{ij}=1)\, P(\delta(i,j) \mid M_{ij}=1)$

$P(M_{ij}=0, \delta(i,j)) = P(M_{ij}=0)\, P(\delta(i,j) \mid M_{ij}=0)$

$$P(\delta(i,j) \mid M_{ij}=0) = \prod_{k=1}^{K} b_k^{\delta_k(i,j)} (1-b_k)^{1-\delta_k(i,j)}$$

$$P(M_{ij}=m, \delta(i,j)) = \left[p \cdot \prod_{k=1}^{K} a_k^{\delta_k(i,j)} (1-a_k)^{1-\delta_k(i,j)} \right]^{m} \times \left[(1-p) \prod_{k=1}^{K} b_k^{\delta_k} (1-b_k)^{1-\delta_k} \right]^{1-m}$$

$$L(\Theta) = \prod_{r=1}^{n} \left[p \cdot \prod_{k=1}^{K} a_k^{\delta_k(i,j)} (1-a_k)^{1-\delta_k(i,j)} \right]^{M_r} \times \left[(1-p) \prod_{k=1}^{K} b_k^{\delta_k} (1-b_k)^{1-\delta_k} \right]^{1-M_r}$$

$$l_c(\Theta) = \sum_{r=1}^{n} \Big[M_r \cdot \log p + (1-M_r) \log(1-p) + M_r \sum_{k=1}^{K} [\delta_{rk} \log a_k + (1-\delta_{rk}) \log(1-a_k)] + (1-M_r) \sum_{k=1}^{K} [\delta_{rk} \log b_k + (1-\delta_{rk}) \log(1-b_k)] \Big]$$

E-step.

$\gamma_r = P(M_r = 1 \mid \delta_r, \Theta^{(t)})$

$$\gamma_r = \frac{P(\delta_r \mid M_r=1, \Theta^{(t)})\, P(M_r=1 \mid \Theta^{(t)})}{P(\delta_r \mid \Theta^{(t)})}$$

$P(\delta_r \mid M_r=1)\, P(M_r=1) + P(\delta_r \mid M_r=0)\, P(M_r=0)$

$$\gamma_r^{(t)} = \frac{p^{(t)} A_r^{(t)}}{p^{(t)} A_r^{(t)} + (1-p^{(t)}) B_r^{(t)}}$$

$$A_r^{(t)} = \prod_{k=1}^{K} (a_k^{(t)})^{\delta_{rk}} (1-a_k^{(t)})^{1-\delta_{rk}}$$

$$B_r^{(t)} = \prod_{k=1}^{K} (b_k^{(t)})^{\delta_{rk}} (1-b_k^{(t)})^{1-\delta_{rk}}$$

$$Q(\Theta \mid \Theta^{(t)}) = \sum_{r=1}^{n} \left[\gamma_r^{(t)} \cdot \log p + (1-\gamma_r^{(t)}) \log(1-p) + \gamma_r^{(t)} \sum_{r=1}^{n} [\delta_{rk} \log a_k + (1-\delta_{rk}) \log(1-a_k)] \right.$$
$$\left. + (1-\gamma_r^{(t)}) \sum_{r=1}^{n} [\delta_{rk} \log b_k + (1-\delta_{rk}) \log(1-b_k)] \right]$$

M-step

① $$Q_p = \sum_{r=1}^{n} [\gamma_r^{(t)} \log p + (1-\gamma_r^{(t)}) \log(1-p)]$$

$$\frac{\partial Q_p}{\partial p} = \sum_{r=1}^{n} \left[\frac{\gamma_r^{(t)}}{p} - \frac{(1-\gamma_r^{(t)})}{1-p} \right] = 0$$

$$\sum_{r=1}^{n} [\gamma_r^{(t)}(1-p) - (1-\gamma_r^{(t)}) p] = 0$$

$$\sum_{r=1}^{n} [\gamma_r^{(t)} - \gamma_r^{(t)} p - p + \gamma_r^{(t)} p] = 0$$

$$\sum_{r=1}^{n} \gamma_r^{(t)} - \sum_{r=1}^{n} p = 0$$

$$mp = \sum_{r=1}^{n} \gamma_r^{(t)}$$

$$p = \frac{1}{m} \sum_{r=1}^{n} \gamma_r^{(t)}$$

② $$Q_{ak} = \sum_{r=1}^{n} \gamma_r^{(t)} [\delta_{rk} \log a_k + (1-\delta_{rk}) \log(1-a_k)]$$

$$\frac{\partial Q_{ak}}{\partial a_k} = \sum_{r=1}^{n} \gamma_r^{(t)} \left[\frac{\delta_{rk}}{a_k} - \frac{1-\delta_{rk}}{1-a_k} \right] = 0$$

$$\sum_{r=1}^{n} \gamma_r^{(t)} [\delta_{rk}(1-a_k) - a_k(1-\delta_{rk})] = 0$$

$$\sum_{r=1}^{n} \gamma_r^{(t)} \delta_{rk} - \sum_{r=1}^{n} \gamma_r^{(t)} a_k = 0$$

$$a_k \sum_{r=1}^{n} \gamma_r^{(t)} = \sum_{r=1}^{n} \gamma_r^{(t)} f_{rk}$$

$$a_k^{(t+1)} = \frac{\sum_{r=1}^{n} \gamma_r^{(t)} f_{rk}}{\sum_{r=1}^{n} \gamma_r^{(t)}}$$

③ $Q_{b_k} = \sum_{r=1}^{n} (1-\gamma_r^{(t)}) [f_{rk} \log b_k + (1-f_{rk}) \log (1-b_k)]$

$$\sum_{r=1}^{n} (1-\gamma_r^{(t)}) f_{rk} - b_k \sum_{r=1}^{n} (1-\gamma_r^{(t)}) = 0$$

$$b_k \sum_{r=1}^{n} (1-\gamma_r^{(t)}) = \sum_{r=1}^{n} (1-\gamma_r^{(t)}) f_{rk}$$

$$b_k^{(t+1)} = \frac{\sum_{r=1}^{n} (1-\gamma_r^{(t)}) f_{rk}}{\sum_{r=1}^{n} (1-\gamma_r^{(t)})}$$

4.

$$X_i \sim \begin{cases} N(\mu_1, \sigma_1^2) & z_i = 1 \\ N(\mu_2, \sigma_2^2) & z_i = 0 \end{cases}$$

$z_i \sim \text{Bernoulli}(\pi)$

$\pi = P(z_i = 1)$

$X_i \mid z_i = 1 \sim N(\mu_1, \sigma_1^2)$

$X_i \mid z_i = 0 \sim N(\mu_2, \sigma_2^2)$

$\Theta = (\pi, \mu_1, \mu_2, \sigma_1^2, \sigma_2^2)$

$P(z_i, x_i \mid \Theta) = P(z_i \mid \Theta) P(x_i \mid z_i, \Theta)$ $\qquad f_1(x_i) = f_0(x_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$

$$= \pi^{z_i} (1-\pi)^{1-z_i} \cdot f_1(x_i)^{z_i} f_0(x_i)^{1-z_i}$$

$$Lc(\Theta) = \prod_{i=1}^{m} \left[\pi^{z_i} (1-\pi)^{1-z_i} \cdot f_1(x_i)^{z_i} f_0(x_i)^{1-z_i}\right]$$

$$\log(\Theta) = \sum_{i=1}^{n} [z_i \log \pi + (1-z_i) \log (1-\pi) + z_i \log f_1(x_i) + (1-z_i) \log f_0(x_i)$$

z_i is unknown.

$$\gamma_i^{(t)} = P(z_i = 1 \mid x_i = x_i, \Theta^{(t)})$$

$$\gamma_i^{(t)} = \frac{\pi^{(t)} f_1(x_i; \Theta^{(t)})}{\pi^{(t)} f_1(x_i; \Theta^{(t)}) + (1-\pi^{(t)}) f_0(x_i; \Theta^{(t)})}$$

$$w_i^{(t)} = \begin{cases} z_i & i = 1, 2, \cdots, m \\ \gamma_i^{(t)} & i = m+1, m+2, \cdots, n \end{cases}$$

$$\log(\Theta) = \sum_{i=1}^{n} \left[w_i^{(t)} \log \pi + (1-w_i^{(t)}) \log(1-\pi) + w_i^{(t)} \log f_1(x_i) + (1-w_i^{(t)}) \log f_0(x_i) \right]$$

$$Q\pi = \sum_{i=1}^{n} \left[w_i^{(t)} \log \pi + (1-w_i^{(t)}) \log(1-\pi) \right]$$

$$\frac{\partial Q\pi}{\partial \pi} = \sum_{i=1}^{n} \left[\frac{w_i^{(t)}}{\pi} - \frac{(1-w_i^{(t)})}{1-\pi} \right] = 0$$

$$\sum_{i=1}^{n} \left[(1-\pi) w_i^{(t)} - \pi (1-w_i^{(t)}) \right] = 0$$

$$\sum_{i=1}^{n} \left[w_i^{(t)} - \pi w_i^{(t)} - \pi + \pi w_i^{(t)} \right] = 0$$

$$\sum_{i=1}^{n} w_i^{(t)} = n\pi$$

$$\pi^{(t+1)} = \frac{1}{n} \sum_{i=1}^{m} w_i^{(t)}$$

$$Q\mu_1 = \sum_{i=1}^{n} w_i^{(t)} \log f_1(x_i)$$

$$f_1(x_i) = \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left(- \frac{(x_i-\mu_1)^2}{2\sigma_1^2} \right)$$

$$\log f_1(x_i) = -\frac{1}{2} \log(2\pi\sigma_i^2) - \frac{(x_i-\mu_1)^2}{2\sigma_i^2}$$

$$Q\mu_1 = \sum_{i=1}^{n} w_i^{(t)} \cdot \left[-\frac{1}{2} \log(2\pi\sigma_i^2) - \frac{(x_i-\mu_1)^2}{2\sigma_1^2} \right]$$

$$\frac{\partial Q\mu_1}{\partial \mu_1} = \sum w_i^{(t)} \frac{1}{2\sigma_1^2} \cdot 2(x_i-\mu_1) \cdot (-1) = \sum w_i^{(t)} \frac{x_i-\mu_1}{\sigma_1^2} = 0$$

$$\sum_{i=1}^{n} w_i^{(t)} \frac{x_i - \mu_1}{\sigma_1^2} = 0$$

$$\sum_{i=1}^{n} w_i^{(t)} x_i - \mu_1 \sum_{i=1}^{n} w_i^{(t)} = 0$$

$$\mu_1 \sum_{i=1}^{n} w_i^{(t)} = \sum_{i=1}^{n} w_i^{(t)} x_i$$

$$\mu_1 = \frac{\sum_{i=1}^{n} w_i^{(t)} x_i}{\sum_{i=1}^{n} w_i^{(t)}}$$

$$Q_{\mu_2} = (1 - w_i^{(t)}) [\log f_0(x_i)]$$

Let's be $1 - w_i^{(t)} = \gamma_i^{(t)}$

Then the equation structure is identical to Q_{μ_1}.

Thus

$$\mu_2 = \frac{\sum_{i=1}^{n} (1 - w_i^{(t)}) x_i}{\sum_{i=1}^{n} (1 - w_i^{(t)})}$$

$$Q_{\sigma_1^2} = \sum_{i=1}^{n} w_i^{(t)} \left[-\frac{1}{2} \log(2\pi\sigma_1^2) - \frac{(x_i - \mu_1)^2}{2\sigma_1^2} \right]$$

$\sigma_1^2 = V$

$$\frac{\partial Q_V}{\partial V} \left[-\frac{1}{2} \log(2\pi V) \right] = \frac{\partial Q_V}{\partial V} \left[-\frac{1}{2} \log(2\pi) - \frac{1}{2} \log(V) \right]$$

$$= 0 - \frac{1}{2} \cdot \frac{1}{V}$$

$$\frac{\partial Q_V}{\partial V} \left[-\frac{(x_i - \mu_1)^2}{2V} \right] = \frac{(x_i - \mu_1)^2}{2V^2}.$$

$$\sum_{i=1}^{n} w_i^{(t)} \left[-\frac{1}{2V} + \frac{(x_i - \mu_1)^2}{2V^2} \right] = 0$$

$$\sum_{i=1}^{n} w_i^{(t)} [-V + (x_i - \mu_1)^2] = 0$$

$$-V \sum_{i=1}^{n} w_i^{(t)} = -\sum_{i=1}^{n} w_i^{(t)} (x_i - \mu_1)^2$$

$$V = \frac{\sum_{i=1}^{n} w_i^{(t)} (x_i - \mu_1)^2}{\sum_{i=1}^{n} w_i^{(t)}}$$

σ_1^2 ↗

$$\sigma_2^2 = \frac{\sum_{i=1}^{n} (1 - w_i^{(t)}) (x_i - \mu_2)^2}{\sum_{i=1}^{n} (1 - w_i^{(t)})}$$